

Inverse-Weight Lin–Lu–Yau Curvature on a Weighted Triangle: Exact Interior Fixed Points and a Nonuniform Stationary Metric

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Abstract

We consider the normalized continuous Lin–Lu–Yau Ricci flow on the weighted triangle when the transition probability along an incident edge is proportional to the inverse of its raw length, while transport cost and the curvature denominator use the induced shortest-path metric. Exact primal and dual Kantorovich certificates give the curvature vector in the central triangle-inequality chamber and in all three one-long-edge chambers, with componentwise agreement on their walls. We then classify every positive normalized fixed point. Besides the uniform point, the fixed set consists of all permutations of

$$\left(\frac{r}{r+2}, \frac{1}{r+2}, \frac{1}{r+2} \right), \quad r \geq 2.$$

In particular, $(2/3, 1/6, 1/6)$ is a strict-chamber stationary metric, with all three curvatures equal to $6/5$. Its omega-limit set is its own singleton, so convergence to the uniform metric does not hold for every positive initial metric. We do not classify omega limits of nonstationary initial data or boundary degenerations; that global phase-portrait problem remains unresolved.

1 Context and precise model

Bai and Hua study the inverse-weight Lin–Lu–Yau flow on finite trees and establish curvature-convergence and convex-gradient-flow results in that setting [2]. Their cyclic case is not covered. The general weighted-graph framework of Bai, Lin, Lu, Wang, and Yau supplies the shortest-path curvature denominator and the normalized continuous-flow convention [3]. A static calculation for a triangle also appears in Bai’s work on unicyclic graphs [1], but it uses a uniform-neighbor transition kernel, rather than the inverse-weight kernel used here. Thus that calculation agrees with the present kernel only at the uniform metric and does not determine the fixed set below.

The exact algebraic treatment of exceptional parameter strata in a graph-indexed polynomial dynamical system has a methodological analogue in Zeng’s work on Lotka–Volterra tree systems [4]. That paper concerns neither optimal transport nor discrete Ricci curvature; we cite it only for this limited methodological comparison.

Label the vertices of C_3 by 1, 2, 3 and the opposite edge lengths by

$$a = w_{23}, \quad b = w_{31}, \quad c = w_{12}.$$

Throughout, $a, b, c > 0$. The induced endpoint distances are

$$d_a = \min\{a, b + c\}, \quad d_b = \min\{b, c + a\}, \quad d_c = \min\{c, a + b\}.$$

At a vertex x , let

$$D_x = \sum_{z \sim x} w_{xz}^{-1}.$$

For $0 \leq \alpha < 1$, the lazy inverse-weight measure based at x is

$$m_x^\alpha(x) = \alpha, \quad m_x^\alpha(z) = (1 - \alpha) \frac{w_{xz}^{-1}}{D_x} \quad (z \sim x).$$

Writing W_1 for Wasserstein distance with cost equal to the induced path metric, define for an edge $e = xy$

$$\kappa_e = \lim_{\alpha \uparrow 1} \frac{1 - W_1(m_x^\alpha, m_y^\alpha)/d(x, y)}{1 - \alpha}. \quad (1)$$

It is essential here that the denominator is $d(x, y)$, not necessarily the raw edge length: in a long-edge chamber the raw edge is not a shortest route between its endpoints.

Common scaling of (a, b, c) changes all transport costs and distances by the same factor and leaves the curvatures unchanged. We therefore normalize $a + b + c = 1$ for the flow and set

$$K = a\kappa_a + b\kappa_b + c\kappa_c, \quad \dot{w}_e = -w_e(\kappa_e - K). \quad (2)$$

At a positive normalized metric, (2) vanishes if and only if all three edge curvatures are equal.

2 An exact three-point transport certificate

The following calculation removes any need to infer curvature from a numerical transport program.

Lemma 1 (Three-point certificate). *Let x, y, z be a three-point metric space with*

$$D = d(x, y), \quad X = d(x, z), \quad Y = d(y, z).$$

Suppose the non-lazy transition probabilities from x to y, z are $p, 1 - p$, while those from y to x, z are $q, 1 - q$. Put $A = q - p$ and $\varepsilon = 1 - \alpha$. For every sufficiently small $\varepsilon > 0$,

$$W_1(m_x^{1-\varepsilon}, m_y^{1-\varepsilon}) = \begin{cases} D[1 - \varepsilon(1 + q)] + \varepsilon AY, & A \geq 0, \\ D[1 - \varepsilon(1 + p)] - \varepsilon AX, & A \leq 0. \end{cases} \quad (3)$$

Consequently,

$$\kappa(x, y) = \begin{cases} 1 + q - AY/D, & A \geq 0, \\ 1 + p + AX/D, & A \leq 0. \end{cases} \quad (4)$$

Proof. The signed surplus $m_x^{1-\varepsilon} - m_y^{1-\varepsilon}$ at x, y, z is

$$1 - \varepsilon(1 + q), \quad -1 + \varepsilon(1 + p), \quad \varepsilon A.$$

For small positive ε , the first entry is positive and the second is negative. If $A \geq 0$, send the surplus εA from z to y and all the surplus at x to y . This plan has the first cost in (3). The dual potential

$$f(y) = 0, \quad f(x) = D, \quad f(z) = Y$$

has the same objective value. It is 1-Lipschitz because $|D - Y| \leq X$.

If $A \leq 0$, send $-\varepsilon A$ from x to z and the remaining required mass from x to y . Its cost is the second expression in (3). The matching dual potential is

$$f(y) = 0, \quad f(x) = D, \quad f(z) = D - X,$$

which is 1-Lipschitz because $|D - X| \leq Y$. Primal and dual values agree in both cases. Dividing the exact affine deficit in (3) by εD proves (4). At $A = 0$ the two formulas coincide. $\square$

For example, orient edge $a = 23$ from 2 to 3 and take $z = 1$. Directly from the inverse-weight kernel,

$$p = \frac{c}{a + c}, \quad q = \frac{b}{a + b}, \quad A = \frac{a(b - c)}{(a + b)(a + c)}, \quad (5)$$

and $(D, X, Y) = (d_a, d_c, d_b)$. Cyclic permutation gives the other edges.

3 The four chamber formulas

The positive cone is covered by the central triangle-inequality chamber and the three chambers in which one raw edge is at least the sum of the other two.

Proposition 2 (Exact curvature vector). *In the central chamber, let e be one raw edge length and let $M \geq m$ be the other two. Then*

$$\kappa_e = \frac{e^2 + 2eM + em + 3Mm - M^2}{(e + M)(e + m)}. \quad (6)$$

If instead $L \geq u + v$ is a long raw edge and $S = L + u + v$, then

$$\kappa_L = \frac{S[L(u + v) + 2uv]}{(L + u)(L + v)(u + v)}, \quad \kappa_u = \frac{S}{L + u}, \quad \kappa_v = \frac{S}{L + v}. \quad (7)$$

The formulas agree componentwise on $L = u + v$.

Proof. Substitute (5) and its cyclic permutations into (4). In the central chamber the metric distances are the corresponding raw lengths. If, for instance, $b \geq c$, then

$$\kappa_a = 1 + \frac{b}{a + b} - \frac{b(b - c)}{(a + b)(a + c)},$$

which simplifies to (6); swapping b, c gives the same symmetric expression. The two remaining components follow cyclically.

In the chamber $L \geq u + v$, the endpoint distance across the long edge is $u + v$, while the short edges remain geodesic. The same substitution into (4) and collection of denominators gives (7).

For completeness, suppose $u \geq v$ and set $L = u + v$. Both formulas give for the long component

$$\frac{2(u^2 + 4uv + v^2)}{(2u + v)(u + 2v)}.$$

For the u component they both give $2(u + v)/(2u + v)$, and for the v component they both give $2(u + v)/(u + 2v)$. Hence agreement holds on the wall; the other walls follow by permutation. $\square$

4 Complete positive fixed-point classification

Theorem 3. *On the normalized open simplex $a + b + c = 1$, the complete fixed-point set of (2) is*

$$\{(1/3, 1/3, 1/3)\} \cup \bigcup_{\sigma \in S_3} \left\{ \sigma \left(\frac{r}{r + 2}, \frac{1}{r + 2}, \frac{1}{r + 2} \right) : r \geq 2 \right\}. \quad (8)$$

Repeated permutations in (8) represent the same three one-parameter branches.

Proof. Sort the raw lengths as $L \geq M \geq m > 0$. First suppose $L \leq M + m$. Subtraction in (6) gives

$$\kappa_M - \kappa_m = -\frac{2L(M - m)}{(L + M)(L + m)}. \quad (9)$$

Thus equal curvature forces $M = m = x$. With this specialization,

$$\kappa_L = \frac{L + 2x}{L + x}, \quad \kappa_x = \frac{2x^2 + 5Lx - L^2}{2x(L + x)},$$

and therefore

$$\kappa_L - \kappa_x = \frac{(L - x)(L - 2x)}{2x(L + x)}. \quad (10)$$

The chamber inequalities give $x \leq L \leq 2x$, so equality occurs only when $L = x$ or $L = 2x$. These cases yield the uniform point and the three wall metrics of ratio 2 : 1 : 1.

Now suppose $L \geq M + m$. Formula (7) gives

$$\kappa_M = \frac{S}{L + M}, \quad \kappa_m = \frac{S}{L + m}.$$

Equal curvature again forces $M = m = x$. Conversely, for every $L \geq 2x$, (7) reduces to

$$\kappa_L = \frac{S}{L + x} = \kappa_x.$$

Thus all and only the ratios $(r, 1, 1)$ with $r \geq 2$, together with the uniform ratio, are fixed. Permutation and normalization give (8). $\square$

Corollary 4 (Failure of universal convergence). *For the positive normalized initial metric*

$$w^0 = (a, b, c) = (2/3, 1/6, 1/6),$$

the solution of (2) is constant and

$$\omega(w^0) = \{(2/3, 1/6, 1/6)\}.$$

In particular, not every positive initial metric converges to $(1/3, 1/3, 1/3)$.

Proof. The ratio is $(4, 1, 1)$, strictly inside a long-edge chamber. Formula (7) gives $\kappa_a = \kappa_b = \kappa_c = 6/5$. Because $a + b + c = 1$, also $K = 6/5$, so the vector field in (2) vanishes. It is a smooth rational vector field near this strict-chamber point; local uniqueness therefore prevents a solution from departing from the constant one. The constant solution exists for all $t \geq 0$ and has the stated singleton omega-limit set. $\square$

5 Scope, reproducibility, and assistance disclosure

The theorem answers the interior-fixed-point part of the triangle problem and the corollary decides the universal uniform-convergence subclaim. It does not give basins or stability types for the nonuniform fixed branches, prove a global wall-crossing theorem for arbitrary nonstationary data, classify their omega-limit sets, or describe boundary rates. At a zero edge, the inverse-weight transition rule is singular, so no boundary ODE or invariant face is asserted here.

The accompanying Python file `evidence/check_curvature.py` performs exact rational-arithmetic checks of (4), (6), (7), wall agreement, and the fixed-point criterion on all $24^3 = 13,824$ triples in $\{1, \dots, 24\}^3$. It finds 456 fixed triples and separately checks the ratio $(4, 1, 1)$. This finite calculation is a reproducibility check only; the primal–dual certificates and the algebraic proof above establish the stated results.

Artificial-intelligence assistance was used in the algebraic exploration, exact-checker preparation, and manuscript drafting. The mathematical claims retained here are limited to the exact statements proved in the text, and no floating-point trajectory, automated verdict, or AI output is treated as a proof. Bibliographic comparisons are likewise stated only at the scope supported by the cited records.

References

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